

Tai Hyoung (William) Rhee

EDUCATION

Seoul National University, Republic of Korea

Mar. 2025 - Present

- M.S. student in Mechanical Engineering
- Advised by Dr. Ayoung Kim

Seoul National University, Republic of Korea

Sep. 2018 - Feb. 2025

- B.S. in Mechanical Engineering
- Graduated *cum laude*

PUBLICATIONS

International Journal

1. Seungsang Yun, Hyunsoo Jang, Tai Hyoung Rhee, Hyunho Song, Hyeonjae Gil, and Ayoung Kim. Cross-spectral stereo inertial odometry. *IEEE Robotics and Automation Letters*, 2026. Accepted, to appear

International Conference Proceedings

1. Dong-Guw Lee*, Tai Hyoung Rhee*, Hyunsoo Jang, Young-Sik Shin, Ukcheol Shin, and Ayoung Kim. Thera: Thermal-aware visual-language prompting for controllable rgb-to-thermal infrared translation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026
2. Tai Hyoung Rhee, Dong guw Lee, and Ayoung Kim. Tidy: Thermal infrared image denoising via wavelet domain entropy and directional stripe index. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026
3. Jeongho Noh, Tai Hyoung Rhee, Eunho Lee, Jeongyun Kim, Sunwoo Lee, and Ayoung Kim. Clutt3r-seg: Sparse-view 3d instance segmentation for language-grounded grasping in cluttered scenes. *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2026

Workshop

1. Tai Hyoung Rhee, Dong guw Lee, and Ayoung Kim. Tir-diffusion: Diffusion-based thermal infrared image denoising via latent and wavelet domain optimization. *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA) Workshops*, 2025

REVIEWER

- IEEE Robotics and Automation Letters (RA-L)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE International Conference on Intelligent Robots and Systems (IROS)

AWARDS AND HONORS

2025 “Graduated *Cum Laude*”, ME, SNU

LANGUAGES & SKILLS

- English (Native), Korean (Native), Mandarin (Conversational)
- MATLAB, C/C++, Python, ROS, Pytorch, Docker, Javascript, CSS, HTML

- L^AT_EX, Microsoft Office, AutoCAD, SolidWorks, Ubuntu, Windows

RESEARCH INTEREST

- Sensor Fusion for SLAM
- Visual Inertial Odometry
- Thermal Infrared Imaging